School of Electrical Engineering and Computer Science CS 362 Software Engineering II

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EdgeGuard Hybrid Intelligence

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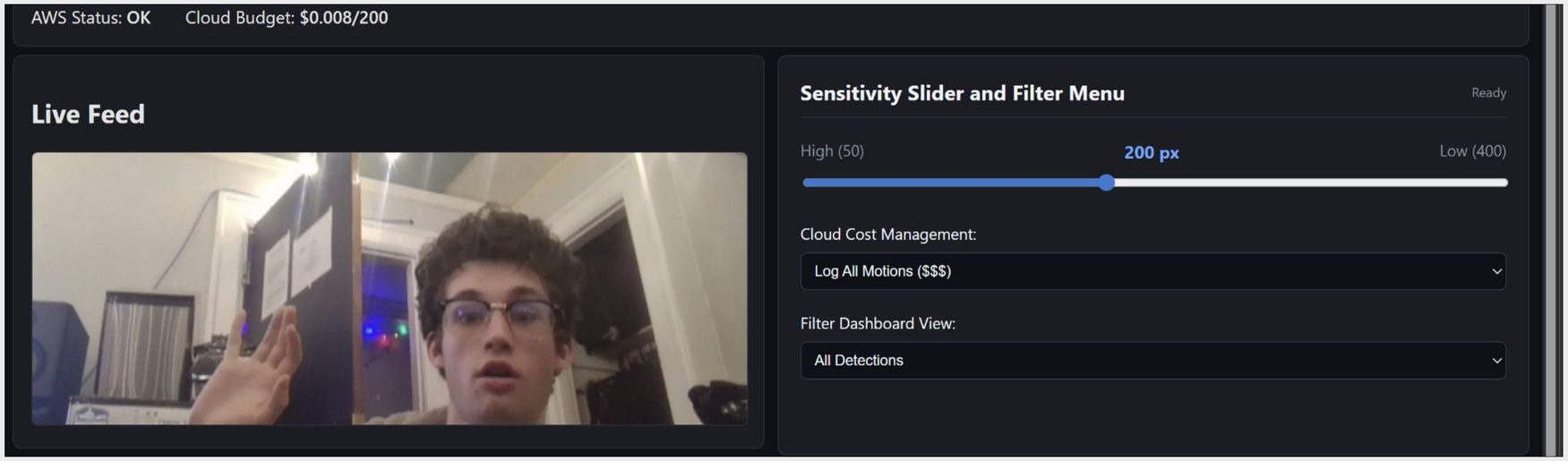


Abstract

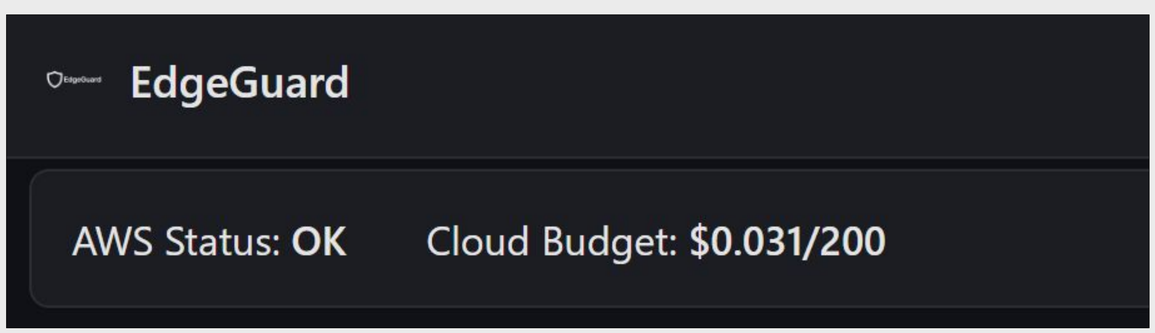
The Problem: Most surveillance systems rely on continuous cloud streaming, leading to massive storage costs and "noise"

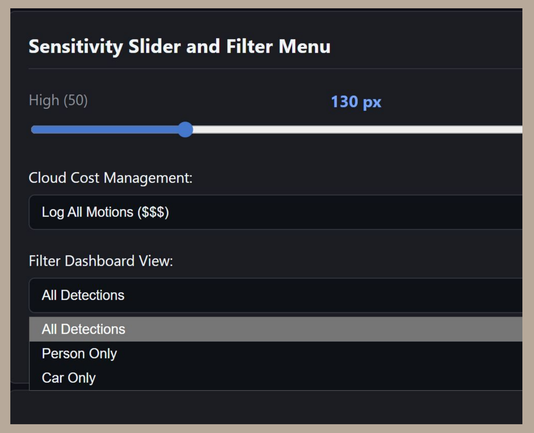
The Solution: EdgeGuard shifts decision-making to the local device. By filtering motion at the edge, we reduce cloud API pings and storage overhead by up to 90%

The Result: Enterprise-grade security for homeowners and small businesses without the high-end price tag

Main Product Features

- **Hybrid Filtering:** Local OpenCV motion detection ignores up to 90% of environmental noise based on the settings the users selected sensitive
- **Intelligent Alerting:** Integrates with Amazon Rekognition to identify specific objects (people, animals, cars)
- **Event-Driven Savings:** Optimized serverless architecture delivers AI intelligence at a fraction of the cost





Technical Approach

- **Edge Intelligence:** Filters up to 90% of noise locally to ensure only real events are recorded
- **Smart Cloud Sync:** Automatically saves high-definition evidence to the cloud when motion is detected.
- **AI Recognition:** Uses enterprise-grade vision to identify people, animals, and vehicles instantly
- **Real-Time Alerts:** A fast, responsive dashboard to monitor your security from any device

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